LATIHAN LOGIC GATES AND COMBINATIONAL CIRCUITS

DDAK 2022/2023

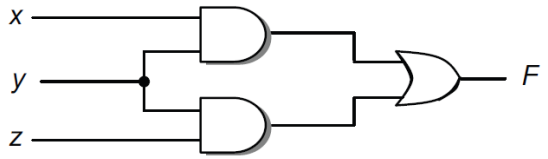
By: Heri Kurniawan (Rev: 28-Feb-2023)

PART I

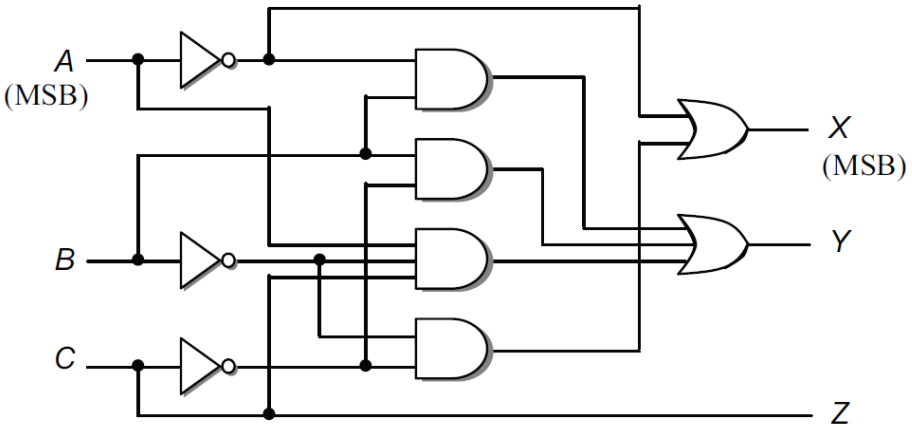
4-1. Draw a logic circuit for the expression $B{\cdot}D + B{\cdot}E + D'{\cdot}F$.

4-2. Draw a logic circuit for the expression $A'{\cdot}B{\cdot}C + B'{\cdot}C{\cdot}D + B{\cdot}C'{\cdot}D + A{\cdot}B{\cdot}D'$.

4-3. What is the Boolean expression for F in the logic circuit below? Express it in SOP form.



4-4. What is the function of the circuit below?



a. Convert from 3-bit sign-magnitude code to 3-bit excess-4 code
b. Convert from 3-bit excess-4 code to 3-bit sign-magnitude code
c. Convert from 3-bit 2's complement number to 3-bit excess-4 code
d. Convert from 3-bit excess-4 code to 3-bit 2's complement number
e. Convert from 3-bit 2's complement number to 1's complement number

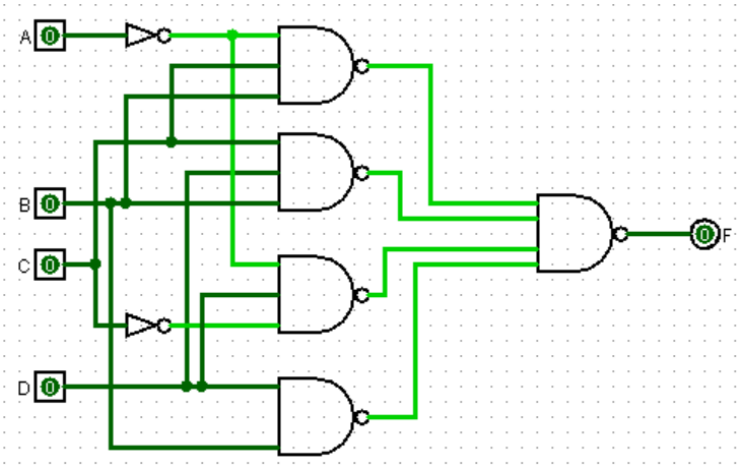
4-6. Prove that the set {AND, NOT} is a complete set of logic gates.

4-7. Draw a logic circuit for $B{\cdot}D+B{\cdot}E+D'{\cdot}F$ using only NAND gates (for the inverters as well).

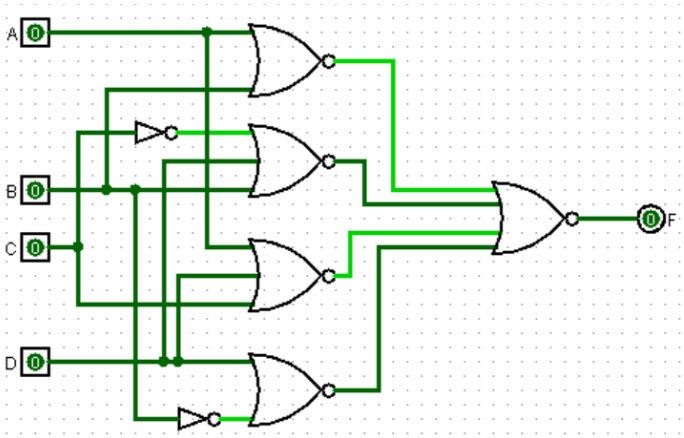
4-8. Draw a logic circuit for $(A+B+C'){\cdot}(B'+C'+D)$ using only NOR gates (for the inverters as well).

PART II

1. Berdasarkan diagram dibawah ini
    a. Cari bentuk persamaan fungsi booleannya
    b. Carilah persamaan SOP-nya
    c. Carilah persamaan SOP optimumnya
    d. Carilah persamaan POS optimumnya.



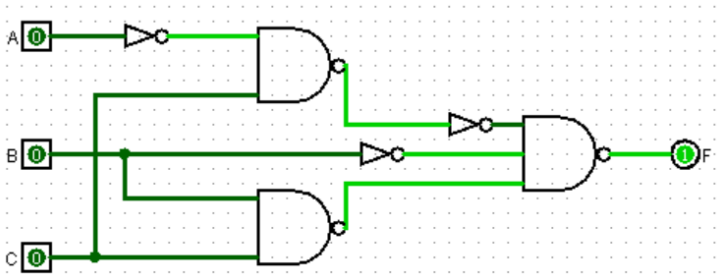
2. Berdasarkan diagram dibawah ini
    a. Cari bentuk persamaan fungsi booleannya
    b. Carilah persamaan POS-nya
    c. Carilah persamaan POS optimumnya
    d. Carilah persamaan SOP optimumnya.



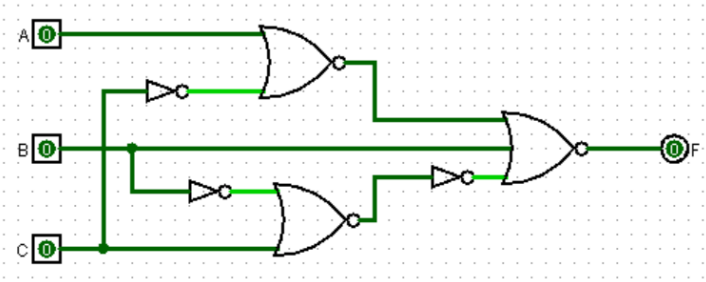
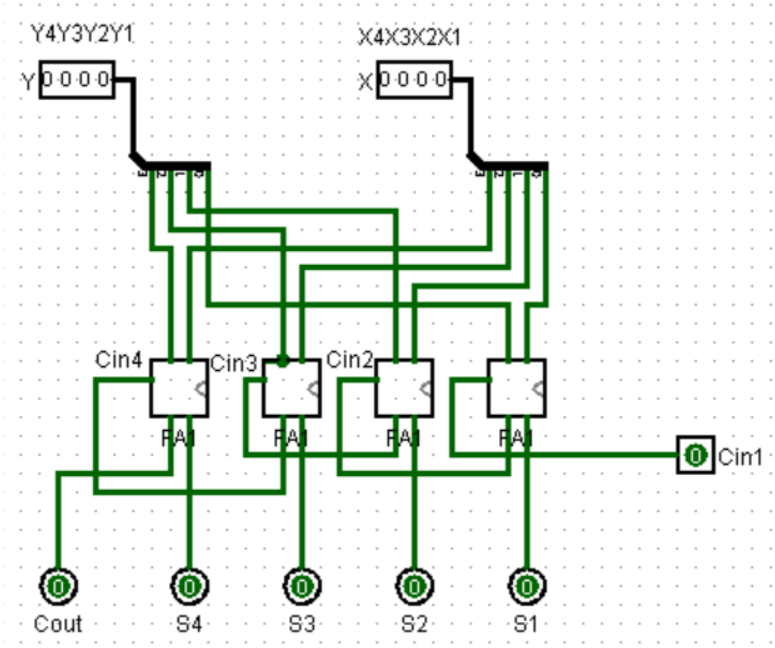
3. Ubahlah F= C'D + AB'C + BD' ke bentuk persamaan yang hanya menggunakan NAND dan Invert

4. Ubahlah F= (A'+B'+C).(B+C').D ke bentuk persamaan yang hanya menggunakan NOR dan Invert

5. Berdasarkan diagram dibawah ini:
    a. Apakah persamaan fungsi booleannya?
    b. Apakah output fungsi F jika input A=1, B=0 dan C=1 ?
    c. Ubahlah circuit hanya dengan gate OR atau Invert



6. Berdasarkan diagram dibawah ini:
    a. Apakah persamaan fungsi booleannya?
    b. Apakah output fungsi F jika input A=0, B=1 dan C=1 ?
    c. Ubahlah circuit hanya dengan gate AND atau Invert



7. Jika input Y4Y3Y2Y1=1011 ; X4X3X2X1=0111; Cin1=1, maka berapakah nilai S4S3S2S1 dan Cout?